

vergence?

in your decision to buy. This is happening now, and seems a good idea to many people.

That said, there is no single definition of convergence. A popular definition on the Internet says that digital convergence is “the technological trend whereby a variety of different digital devices such as televisions, mobile telephones, and now refrigerators are merging into a multi-use communications appliance employing common software to communicate through the Internet.” Someone else says, “What is really happening is the convergence of content, communication, and computing.”

Think about how content reaches us nowadays. Radio, satellite, cable, fibre optics, wireless, the Internet are some of the more common media. This is an example of how our lifestyles are changing, or are being changed, by the fact that we go anywhere we want, do anything we want, and the content reaches us anyway. Cable disconnected? Turn to the Internet. Net connection down? Get it from someone's GPRS phone, on the move. And so on. Content is making its way across boundaries.

Communications is key when we're talking convergence: devices are talking to each other more and more. Your iPod talks to your PC. Your phone talks to another in the sense of being able to transmit a picture to it—whether you're standing near the other phone or whether you're thousands of miles away from it.

Everything has a chip—from phones to cars to stereo systems. And all these chips will talk to each other—or are already talking to each other. Companies want to put a chip into everything.

Says Gottfried Dutiné, an executive vice-president at Philips, “Digitisation is creating products that can't be categorised as tech or consumer electronics. The walls are coming down.” And as an article in *BusinessWeek Online*, dated 21 June 2004, puts it, “That sets up a collision of three massive industries. In one corner stands the \$1.1 trillion computer and software biz, with its American leaders. In another is the \$225 billion consumer-electronics sector, with its strong Asian roots and a host of aggressive new Chinese players. The third camp is the \$2.2 trillion communications industry, a behemoth that extends from wireless powerhouses in Asia and Europe to the networking stars of Silicon Valley. All the three groups will have a hand in building the digital wonders that are headed our way.”

The rush to churn out those digital wonders has reached critical mass. Innovation is now a full-time job. Many products coming out of this rush will flop: this is uncharted territory, and there are few precedents to draw from—there are few standards, and interoperability and security are key issues. The ones that make it will define our digital future.



Some converged gadgets just flop—refer *Crash Landing* in this issue. Others, like the Kaleidescape (at right), are at the cutting edge of innovation

Going back a little, think about the PC. What can't you do on a PC these days? You can have one browser window open showing you the latest cricket scores. Another brings you an e-learning class (you shouldn't be looking at cricket scores when you're studying, but anyway). An MP3 song plays in the background. And on and on... all at the same time, most of them via broadband.

Broadband is the enabler: if communication is to happen and content is to be delivered, you need a fat pipe. How could streaming Internet video be delivered to your smart TV without an always-on broadband connection? The PC, with its broadband connection, is a wonderful example of how 'just a device' became a converged device. Convergence is all around; and with most people owning PCs, the PC had to be affected sooner or later.

As with the refrigerator example, another Holy Grail of convergence is the perfect, networked living room. The all-in-one living room device hasn't yet been manufactured, but it will soon be.

Market factors dominate here: how soon will ultra-high bandwidth get to homes? How soon will large flat-screen TVs become affordable to all? When all that is worked out, the idea is simple enough: one centralised entertainment device with a broadband connection and wireless satellites. The device will be fed from diverse sources, and the satellites may be anywhere—even in your house—and without any wire clutter whatsoever.

In Singapore, Australia and other places, you can pay for a can of Coke at a vending machine with your cell phone—beam your phone at the machine, and your Coke is billed to your phone account. Luddites may ask, why not just pay using coins instead?

But the revolution has already begun. The Coke machine example states what digital convergence is all about: everything digital will talk to everything else digital, and things digital will be near you all the time, everywhere.